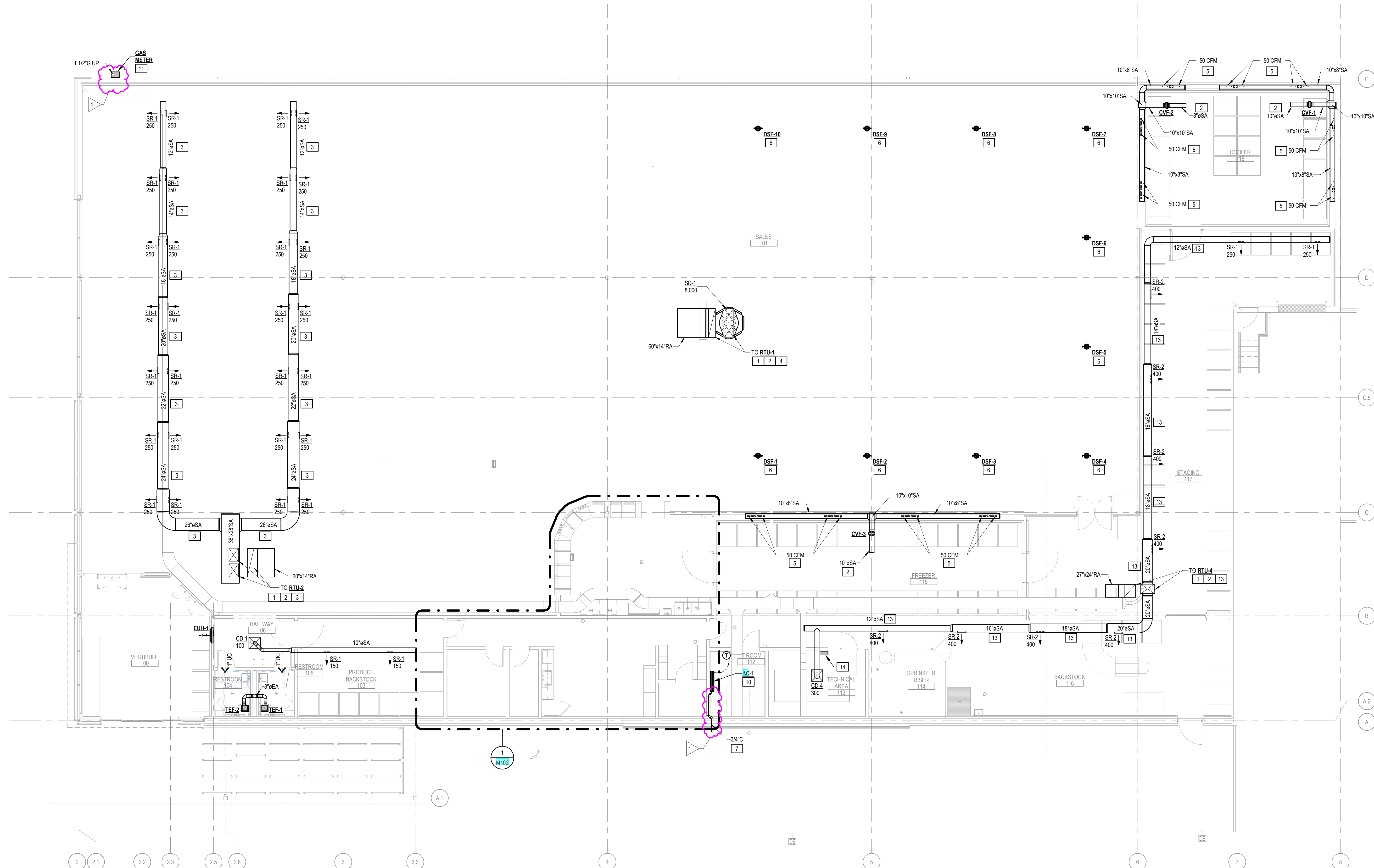


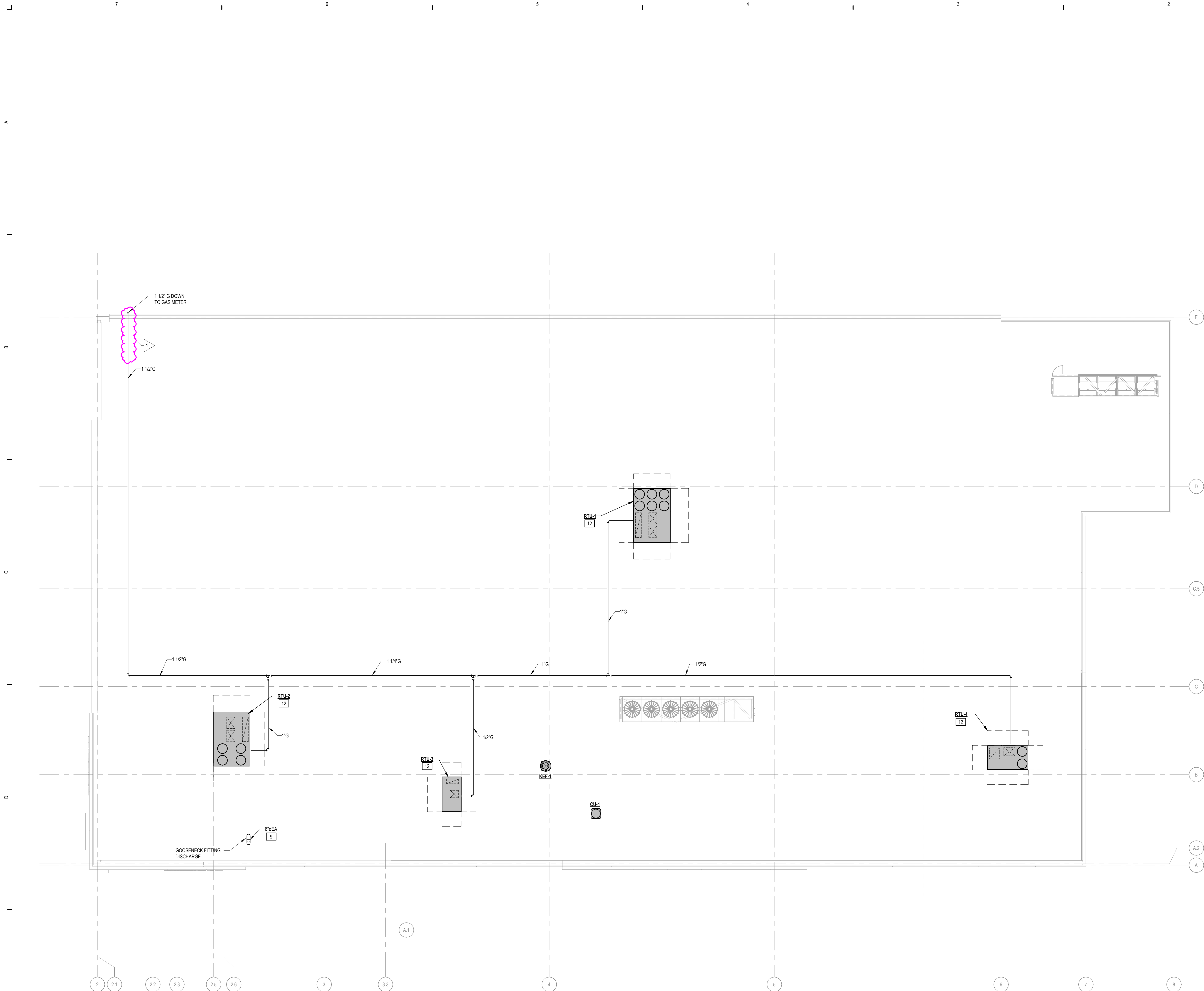


MECHANICAL PLAN

1/8" = 1'-0"

HVAC KEYNOTES

1. DUCT DROPS TO MATCH RUIT CONNECTION SIZE. TRANSITION SIZE OF SHOW UP PLAN. DUCTS DROPS TO BE LINED WITH 1" LINING. TAPE END OF SHOWUP TO SHEET PILE AND COVER WITH 1" LINING.
2. PROVIDE WELDED WIRE MESH SCREEN AT OPEN END OF DUCTWORK. SEE MECHANICAL DETAILS FOR MORE INFORMATION.
3. COORDINATE WITH STRUCTURAL POSITION OF STRUCTURAL JOISTS WHERE POSSIBLE. COORDINATE WITH STRUCTURAL TRADES. COORDINATE DUCT ROUTING TO AVOID CONFLICT WITH STRUCTURAL ELEMENTS. DUCTS TO BE BELOW CABLE TRAY TO AVOID CONFLICT WITH DUCT ROUTING OR AIRFLOW.
4. ADJUST DISCHARGE BLADES IN DRAIN DIFFUSER TO BE 20° FROM HORIZONTAL. 1/2" MIN. CLEARANCE WITH WIRE MESH SCREEN. LOCATED 1" FROM 1/4" A/F. REFER TO MECHANICAL DETAILS.
5. ~~DISCHARGE FROM RAIN PIPING TO BE ADJUSTED AS 40° A/F.~~
6. ~~DUCTWORK TO BE 1/2" MIN. CLEARANCE FROM RAIN DRAIN AND DRAIN TO GRADE. PENETRATION THROUGH EXTERIOR WALL TO BE AT APPROXIMATELY 1/2" A/F.~~
7. ~~EXHAUST DUCT FROM EXHAUST FAN FOR EXHAUST DUCT TO CONNECT TO EXHAUST FAN. EXHAUST DUCT TO BE 1/2" MIN. CLEARANCE FROM EXHAUST FAN.~~
8. EXTEND DISCHARGE OF EXHAUST DUCT TO TOP OF PARAPET AND PROVIDE WITH GOOSENECK.
9. NEW 1" NIPPLE SYSTEM RUIT REFRIGERANT PIPING TO CONDENSING UNIT ON ROOF. SIZE AND MATERIAL OF PIPING TO BE BY MANUFACTURER RECOMMENDATION.
10. GAS METER. SEE NATURAL GAS LOAD SCHEDULE FOR CAPACITY AND PRESSURE INFORMATION. COORDINATE EXACT LOCATION WITH INCOMING GAS SERVICE FROM CURB.
11. PROVIDE AND INSTALL PRESSURE REDUCING STATION TO REDUCE THE NATURAL GAS PRESSURE FROM 2 PHSI TO 7 WIC. PROVIDE AND INSTALL GAS CONNECTION TO NATURAL GAS SUPPLEMENTARY GAS METER. PROVIDE GAS METER WITH GAS EQUIPMENT SCHEDULES AND GAS LOAD. SEE GAS PIPING RUIT CONNECTION WITH PERMITS DATA ON MEMO.
12. ROUTE DUCTWORK AS HIGH AS POSSIBLE. TRYING TO THE BOTTOM OF STRUCTURAL JOISTS. COORDINATE WITH STRUCTURAL TRADES FOR DUCT ELEVATION. DUCT ROUTING TO BE ABOVE CABLE TRAY. DUCT MOUNTED DIFFUSERS TO BE ANGLED 45° FROM VERTICAL.
13. PROVIDE MANUAL VARIABLE SPEED CONTROLLER FOR DISTRICTIONAL FANS. MANUAL VARIABLE SPEED CONTROLLER 1/2" - 120" SAE. INSTALLED BY ELECTRICAL CONTRACTOR. MANUAL VARIABLE SPEED CONTROLLER 1/2" - 40" SAE.



ROOF HVAC PLAN
1/8" = 1'-0"

HVAC KEYNOTES

1. DUCT DROPS TO MATCH RTU CONNECTION SIZE. TRANSITION TO SIZE AS SHOWN ON PLANS. DUCT DROPS TO BE LINED WITH 1" LINING. TAPE END OF DUCTWRAP TO SHEET METAL. EXPOSED CUT END OF INSULATION IS NOT ACCEPTABLE.
2. PROVIDE WELDED WIRE MESH SCREEN AT OPEN END OF DUCTWORK. SEE MECHANICAL DETAILS FOR MORE INFORMATION.
3. ROUTE DUCTS AS HIGH AS POSSIBLE, THROUGH STRUCTURAL JOISTS WHERE POSSIBLE. COORDINATE WITH STRUCTURAL TRADES. COORDINATE DUCT ROUTING WITH CABLE TRAY LOCATION. DUCT ROUTING TO BE BELOW CABLE TRAY TO AVOID CONFLICT WITH DUCT ROUTING OR AIRFLOW.
4. ADJUST DIFFUSER BLADES IN DRUM DIFFUSER TO BE 20 ° FROM HORIZONTAL.
5. 2"x12" SLOT OPENING COVERED WITH WIRE MESH SCREEN, LOCATED 1'-6" AFF. REFER TO MECHANICAL DETAILS.
6. DESTRATIFICATION FAN TO BE MOUNTED AT 18" AFF.
7. ROUTE CONDENSATE DOWN WITHIN WALL. EXIT WALL AND DRAIN TO GRADE. PENETRATION THROUGH EXTERIOR WALL TO BE AT APPROXIMATELY 1'-0" AFF.
8. EXHAUST DUCT FROM EXHAUST FAN ON ROOF. EXHAUST DUCT TO CONNECT TO KITCHEN HOOD. REFER TO BAKERY HOOD DETAILS M102.
9. EXTEND DISCHARGE OF EXHAUST DUCT TO TOP OF PARAPET AND PROVIDE WITH GOOSENECK.
10. NEW SPLIT SYSTEM. ROUTE REFRIGERANT PIPING TO CONDENSING UNIT ON THE ROOF. SIZE AND MATERIAL OF PIPING TO BE BY MANUFACTURER RECOMMENDATION.
11. GAS METER. SEE NATURAL GAS LOAD SCHEDULE FOR CAPACITY AND PRESSURE INFORMATION. COORDINATE EXACT LOCATION WITH INCOMING GAS SERVICE FROM CIVIL.
12. PROVIDE AND INSTALL PRESSURE REDUCING STATION TO REDUCE THE NATURAL GAS PRESSURE FROM 2 PSI TO 7" W.C. PROVIDE AND INSTALL GAS CONNECTION TO GAS FIRED EQUIPMENT WITH DIRT LEG AND GAS COCK. REFER TO M100 FOR EQUIPMENT SCHEDULES AND GAS LOAD. SEE GAS PIPING RTU CONNECTION WITH PRV DETAIL ON M100.
13. ROUTE DUCTS AS HIGH AS POSSIBLE, TIGHT TO THE BOTTOM OF STRUCTURAL JOISTS. COORDINATE WITH STRUCTURAL TRADES FOR DUCT ELEVATION. DUCT ROUTING TO BE ABOVE CABLE TRAY. DUCT MOUNTED DIFFUSERS TO BE ANGLED 45 ° FROM HORIZONTAL.
14. PROVIDE MANUAL VARIABLE SPEED CONTROLLER FOR DESTRATIFICATION FANS. MANUFACTURER AIRUS MODEL TRIAC-120-5. 120V 5AMPS. INSTALLED BY ELECTRICAL CONTRACTOR. INSTALL TIGHT TO EDGE OF WALL AT 4" AFF.

LIDL US, LLC

LIDL 1517 NORCROSS, GA



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ISSUANCE

11/07/2025 PERMIT CYCLE 1
12/18/2025 PERMIT CYCLE 02

PROJECT NUMBER
7960130
PROJECT MANAGER
A. HENGSTEBECK
PROFESSIONAL
M. GROSE
DRAWN BY
J. MARTINEZ
CHECKED BY
B. KOOMAN

ROOF
MECHANICAL
PLAN

M103



FAN SCHEDULE

MARK	LOCATION	BASED ON	MODEL	TYPE	CFM	ESP	SONE	FAN RPM	BACKDRAFT DAMPER REQUIRED	TYPE	HP	V	PH	HZ	FLA	REMARKS
CVF-1	BACKSTOCK FREEZER	FANTECH	FG-10 EC	INLINE BOOSTER	400	0.30		2138	No		0.11	120	1	60	1.27	1,4,8,9,10
CVF-2	BAKERY FREEZER/COOLER	FANTECH	FG-8 EC	INLINE BOOSTER	300	0.30		2131	No		0.06	120	1	60	1.27	1,4,8,9,10
CVF-3	BAKERY FREEZER/COOLER	FANTECH	FG-10 EC	INLINE BOOSTER	400	0.30		2138	No		0.11	120	1	60	1.27	1,4,8,9,10
KEF-1	KITCHEN HOOD	CAPITVAIRE	DU50HFA	CENTRIFUGAL UPBLAST	814	0.50	10.3	1231	No		0.5	120	1	60	6.3	1,2,3,4,5,7
TEF-1	RESTROOM	GREENHECK	SP-A190	CEILING EXHAUST	100	0.40	1.5	887	Yes	GRAVITY	0.16	115	1	60	0.45	1,2,4,6
TEF-2	RESTROOM	GREENHECK	SP-A190	CEILING EXHAUST	100	0.40	1.5	887	Yes	GRAVITY	0.16	115	1	60	0.45	1,2,4,6

1. PROVIDE WITH STANDARD DISCONNECT.
2. PROVIDE WITH GRAVITY BACKDRAFT DAMPER.
3. PROVIDE WITH VFD FOR BALANCING AND SOFT START.
4. PROVIDE WITH VIBRATION ISOLATION.
5. PROVIDE WITH ROOF CURB.
6. FAN OPERATION CONTROLLED VIA BMS BASED ON OCCUPIED/UNOCCUPIED SPACE.
7. FAN TO BE CONTROLLED BY THE KITCHEN HOOD CONTROL PANEL AND BMS.
8. FAN TO BE HORIZONTALLY MOUNTED.
9. PROVIDE WITH MOTOR GUARD ENCLOSURE.
10. FAN OPERATION CONTROLLED VIA BMS AND TO RUN CONTINUOUSLY.

SPLIT SYSTEM AIR CONDITIONING UNIT SCHEDULE

MARK	LOCATIONS	MANUFACTURER	MODEL	COOLING CAPACITY (MBH)	HEATING CAPACITY (MBH)	AIRFLOW (CFM)	DIMENSIONS			UNIT WEIGHT (LBS)	ELECTRICAL				REMARKS
							L	W	H		V	PH	HZ	AMPS	
26-1	IT ROOM 109	LENNOX	MVM0012S6-1P	16	24	340	35	8	11"	23	208 V	1	60 Hz	0.1 A	1,2,3,4

1. PROVIDE WITH WIRED PROGRAMMABLE THERMOSTAT.
2. PROVIDE WITH BACNET-IP GATEWAY.
3. SINGLE POINT POWER CONNECTION. INDOOR UNIT IS POWERED FROM OUTDOOR UNIT.
4. PROVIDE WITH DRIP TRAY. DRIP TRAY TO BE INSTALLED SO BOTTOM OF TRAY IS 7' 0" AFF.

CONDENSING UNIT SCHEDULE

CONDENSING UNIT SCHEDULE													
MARK	MANUFACTURER	MODEL	LOCATION	UNIT SERVED	CAPACITY (TONS)	EER	SEER	ELECTRICAL					REMARKS
								V	PH	HZ	MCA	MOP	
CU-1	LENNOX	MWLD012S6S-1P	ROOF	AG-1	1	14	24	208 V	1	60 Hz	12 A	20 A	1,2

1. MOUNT ON EQUIPMENT RAILS.
2. PROVIDE WITH LOW AMBIENT HEAT PUMP.

ELECTRIC UNIT HEATER SCHEDULE

MARK	BASED ON	MODEL	HEATING LOAD (KW)	ELECTRICAL				WEIGHT (LB)	REMARKS
				FLA	V	PH	HZ		
ELH-1	QMARK	EF080048	8	17 A	480 V	1	60 Hz	50	1,2,3

1. PROVIDE WITH INTEGRAL DISCONNECT.
2. COORDINATE GRILLE COLOR WITH ARCHITECT.
3. PROVIDE WITH INTEGRAL TAMPER PROOF THERMOSTAT SET TO 55 °F.

AIR BALANCE SCHEDULE

TAG	SUPPLY AIR CFM	RETURN AIR CFM	OUTSIDE AIR CFM	EXHAUST AIR CFM	UNIT PRESSURE	SEQUENCE OF OPERATIONS
RTU-1	8000	9600	2400	0	2400	CONTINUOUS
RTU-2	7000	4900	2100	0	2100	CONTINUOUS
RTU-3	2300	1700	600	0	600	CONTINUOUS
RTU-4	4000	3600	400	0	400	CONTINUOUS
KEF-1	0	0	0	814	-814	CONTINUOUS
TEF-1	0	0	0	100	-100	INTERMITTENT
TEF-2	0	0	0	100	-100	INTERMITTENT
TOTAL	21300	15800	5500	1014	4486	POSITIVE

INTERNATIONAL MECHANICAL CODE 2018 VENTILATION FOR ACCEPTABLE INDOOR AIR QUALITY

INTERNATIONAL MECHANICAL CODE 2018 VENTILATION FOR ACCEPTABLE INDOOR AIR QUALITY																
ZONE			403.3.1.1 ZONE OUTDOOR AIRFLOW CALCULATION										DESIGN			
ROOM NUMBER	ROOM NAME	OCCUPANCY CATEGORY (TABLE 403.3.1.1)	OCCUPANT DENSITY (PEOPLE/1000 SF)	PEOPLE OUTDOOR AIRFLOW RATE (CFM/PERSON)	ZONE POPULATION Pz (PEOPLE)	ZONE OUTDOOR AIRFLOW RATE (CFM/Pz)	FLOOR AREA Aa (SF)	BREATHING ZONE OUTDOOR AIRFLOW Vbz=PzRaZz (CFM)	ZONE AIR DISTRIBUTION EFFECTIVENESS Ez	MINIMUM OUTDOOR AIRFLOW Vbz = Vbz/Ez (CFM)	EXHAUST AIRFLOW RATE (CFM/Et)	ZONE SUPPLY AIRFLOW (CFM)	ZONE PRIMARY AIRFLOW Vps (CFM)	ACTUAL OUTDOOR AIRFLOW Voz (CFM)	TOTAL EXHAUST AIRFLOW (CFM)	
RTU-1 AND RTU-2																
101	SALES	SALES	15	7.5	224	0.12	1690	3475.52	0.8	4344.15	0	10000	10000	4000		
					TOTALS		224	1690	3475.52	-	4344.15	-	10000	10000	4000	0
RTU-3																
102	BAKERY	KITCHENP	0	0	0	0	515	0	0.8	0.00	50100P	800	800	200	814	
106	PRODUCE BACKSTOCK	WAREHOUSES	0	10	0	0.08	432	25.92	0.8	32.40	0	300	300	75	0	
108	RESTROOM	TOILET ROOMS- PUBLIC	0	0	0	0	48	0	0.8	0.00	5070P	0	0	0	100	
102	RESTROOM	TOILET ROOMS- PUBLIC	0	0	0	0	48	0	0.8	0.00	5070P	0	0	0	100	
103	HALLWAY	OFFICE SPACES	5	5	1	0.08	117	12.02	0.8	15.03	0	100	100	25	0	
107	OFFICE	OFFICE SPACES	5	5	1	0.08	118	12.08	0.8	15.10	0	100	100	25	0	
108	MEETING ROOM	OFFICE SPACES	5	5	1	0.08	58	8.48	0.8	10.60	0	100	100	25	0	
109	BREAKROOM/HALLWAY	OFFICE SPACES	5	5	1	0.08	266	20.98	0.8	26.20	0	150	150	38	0	
110	BREAKROOM	CONFERENCE/MEETING	50	5	11	0.08	237	19.22	0.8	18.53	0	800	800	150	0	
111	EMPLOYEE CHANGING ROOM	OFFICE SPACES	5	5	1	0.08	48	7.68	0.8	9.85	0	100	100	25	0	
					TOTALS		16	1887	156.56	-	166.7	-	2300	2300	575	1014
RTU-4																
113	TECHNICAL AREA	WAREHOUSES	0	10	0	0.08	493	29.58	0.8	36.98	0	300	300	200	0	
116	BACKSTOCK	WAREHOUSES	0	10	0	0.08	1078	64.56	0.8	80.70	0	1600	1600	200	0	
117	STAGING	WAREHOUSES	0	10	0	0.08	1683	101.98	0.8	128.23	0	2100	2100	200	0	
					TOTALS		0	2759	185.54	-	226.925	-	4000	4000	600	0

NATURAL GAS LOAD

APPLIANCE	TAG	MBH
ROOFTOP UNIT	RTU-1	480
ROOFTOP UNIT	RTU-2	480
ROOFTOP UNIT	RTU-3	150
ROOFTOP UNIT	RTU-4	180
total		1290

NOTES:

1. REFER TO THE 2018 GEORGIA FUEL GAS CODE, TABLE 402.4(5) FOR SIZING INFORMATION.
2. NATURAL GAS SERVICE PRESSURE TO BE 2 PSI.
3. GAS SYSTEM FULLY DEVELOPED LENGTH IS 450 FEET.
4. BLACK STEEL PIPE TO BE USED FOR NATURAL GAS SYSTEM.

ROOF TOP UNIT SCHEDULE

ROOF TOP UNIT SCHEDULE																										
MARK	MANUFACTURER	MODEL	LOCATION	MIN OA	SUPPLY AIR BLOWER				DX COOLING COIL				HEATING MODULE				# OF	EER	WEIGHT	ELECTRICAL				REMARKS		
					CFM	ESP (IN. WC)	MOTOR HP	EDB	EWB	LDB	LWB	TOTAL	SENS.	INPUT	OUTPUT	EDB				TEMP. RISE	STAGES	VOLTS	PHASE		HZ	MCA
RTU-1	LENNOX	LGT240HSM	ROOF	2,400	8,000	0.5	7.50	84 °F	66 °F	64 °F	58 °F	243	182	480	389	51 °F	96 °F	2	12	3147 lb	460 V	3	60 Hz	49 A	50 A	1-21,23
RTU-2	LENNOX	LGT210HSM	ROOF	2,100	7,000	1	5.00	84 °F	66 °F	64 °F	59 °F	221	159	480	389	51 °F	103 °F	2	12	3115 lb	460 V	3	60 Hz	41 A	45 A	1-22
RTU-3	LENNOX	LGT0724HE	ROOF	600	2,300	1	1.50	87 °F	69 °F	66 °F	59 °F	75	56	150	121	53 °F	101 °F	2	12.2	899 lb	460 V	3	60 Hz	16 A	20 A	1-22
RTU-4	LENNOX	LGT120HSE	ROOF	400	4,000	1	3.75	80 °F	64 °F	60 °F	57 °F	126	92	180	144	60 °F	93 °F	2	12.1	1395 lb	460 V	3	60 Hz	23 A	30 A	1-22

1. PROVIDE SINGLE POINT POWER CONNECTION.
2. COORDINATE INLET GAS PRESSURE REGULATOR W/ BUILDING GAS PRESSURE.
3. PROVIDE 14" HEIGHT INSULATED SLOPED ROOF CURB.
4. FACTORY INSTALLED WEATHER PROOF MAIN DISCONNECT.
5. FACTORY INSTALLED PRESSURE SWITCH FOR DIRTY FILTER STATUS.
6. MERV 8 2" PLEATED FILTERS.
7. CERTIFIED IN ACCORDANCE WITH ANSI/AHRI STANDARD 340/360.
8. PROVIDE WITH FACTORY INSTALLED SUPPLY AND RETURN SMOKE DETECTORS.
9. STAINLESS STEEL HEAT EXCHANGER.
10. CONDENSATE DRAIN TRAP.
11. PROVIDED WITH PHASE AND BROWNOUT PROTECTION.
12. PROVIDED TEMPERATURE SENSOR, COORDINATE WITH BMS.
13. PROVIDE DUCT MOUNTED CO2 SENSOR.
14. ENTHALPY CONTROLLED ECONOMIZER.
15. FACTORY INSTALLED 115V GFI OUTLET W/ WEATHERPROOF COVER.
16. PROVIDE WITH AIR INTAKE HOOD.
17. PROVIDED SPACE MOUNTED ON WALL HUMIDITY SENSOR AND HUMIDITY CONTROL.
18. VFD CONTROLLED SUPPLY FAN.
19. HOT GAS REHEAT.
20. HAIL GUARDS PROVIDED BY MFR.
21. INDOOR AND OUTDOOR CORROSION PROTECTION COATING.
22. PROVIDE WITH BAROMETRIC RELIEF.
23. PROVIDE WITH MODULATING POWER EXHAUST.

DESTRATIFICATION FAN SCHEDULE

MARK	MANUFACTURER	MODEL	CFM	ELECTRICAL					REMARKS
				V	PH	HZ	FLA	W	
DSF-1	AIRIUS	PEARL-S1-SP	545	120 V	1	60 Hz	0.32 A	45 W	1,2,3,4,5
DSF-2	AIRIUS	PEARL-S1-SP	545	120 V	1	60 Hz	0.32 A	45 W	1,2,3,4,5
DSF-3	AIRIUS	PEARL-S1-SP	545	120 V	1	60 Hz	0.32 A	45 W	1,2,3,4,5
DSF-4	AIRIUS	PEARL-S1-SP	545	120 V	1	60 Hz	0.32 A	45 W	1,2,3,4,5
DSF-5	AIRIUS	PEARL-S1-SP	545	120 V	1	60 Hz	0.32 A	45 W	1,2,3,4,5
DSF-6	AIRIUS	PEARL-S1-SP	545	120 V	1	60 Hz	0.32 A	45 W	1,2,3,4,5
DSF-7	AIRIUS	PEARL-S1-SP	545	120 V	1	60 Hz	0.32 A	45 W	1,2,3,4,5
DSF-8	AIRIUS	PEARL-S1-SP	545	120 V	1	60 Hz	0.32 A	45 W	1,2,3,4,5
DSF-9	AIRIUS	PEARL-S1-SP	545	120 V	1	60 Hz	0.32 A	45 W	1,2,3,4,5
DSF-10	AIRIUS	PEARL-S1-SP	545	120 V	1	60 Hz	0.32 A	45 W	1,2,3,4,5

1. PROVIDE WITH STANDARD DISCONNECT.
2. PROVIDE WITH SPEED CONTROLLER.
3. DESTRATIFICATION FAN TO BE TIED TO BMS. FAN TO RUN CONTINUOUSLY WHEN OCCUPIED AND TURN OFF WHEN UNOCCUPIED.
4. PROVIDE WITH FAN HANGING KIT.
5. FAN COLOR AND HOUSING TO BE COORDINATED WITH ARCHITECTURAL WHITE UNLESS OTHERWISE NOTED.

DIFFUSER, REGISTER, & GRILLE SCHEDULE

DIFFUSER, REGISTER, & GRILLE SCHEDULE													
MARK	MANUFACTURER	MODEL	CFM RANGE		NECK SIZE		FACE SIZE		MOUNTING	PATTERN	COLOR	REMARKS	
			MIN FLOW	MAX FLOW	Ø"	L"	W"	W"					
CD-1	TITUS	OMNI	0	100	6"		24"	24"	LAY-IN	4-WAY	WHITE	1,2,3,4,5	
CD-2	TITUS	OMNI	101	200	8"		24"	24"	LAY-IN	4-WAY	WHITE	1,2,3,4,5	
CD-3	TITUS	OMNI	201	300	10"		24"	24"	LAY-IN	4-WAY	WHITE	1,2,3,4,5	
CD-4	TITUS	OMNI	301	400	12"		24"	24"	LAY-IN	4-WAY	WHITE	1,2,3,4,5	
SD-1	RUSKIN	01-S78-25	0	8000		36"	36"		DUCT MOUNT	6-WAY	CLEAR ANODIZED	1,2,3,4,5	
SR-1	TITUS	S301FL	0	250	6"	12"	8"	14"	SIDEWALL	DOUBLE DEFLECTION	CLEAR ANODIZED	1,2,3,4,5,6	
SR-2	TITUS	S301FL	251	400	8"	18"	10"	20"	SIDEWALL	DOUBLE DEFLECTION	CLEAR ANODIZED	1,2,3,4,5,6	
CRG-1	TITUS	PAR	0	150	8"		24"	12"	LAY-IN	PERFORATED	WHITE	1,2,3,4,5	
CRG-2	TITUS	PAR	0	150	8"		24"	24"	LAY-IN	PERFORATED	WHITE	1,2,3,4,5	
CRG-3	TITUS	PAR	0	400		12"	12"	24"	LAY-IN	PERFORATED	WHITE	1,2,3,4,5	
RG-1	TITUS	350RL	0	300	10"	10"	12"	12"	SIDEWALL	SINGLE DEFLECTION	WHITE	1,2,3,4,5	

1. PROVIDE WITH MAXIMUM TOTAL PRESSURE DROP OF 0.1" W.G.
2. PROVIDE WITH MAXIMUM SOUND POWER LEVEL OF NC 30.
3. SIZE OF DUCT TAKEOFF, BRANCH DUCT, AND FLEX DUCT TO MATCH THE NECK SIZE OF ASSOCIATED DIFFUSER/GRILLE UNLESS OTHERWISE NOTED.
4. CFM SHOWN ON PLAN.
5. PROVIDE WITH BALANCE DAMPER AT BRANCH TAKEOFF UNLESS OTHERWISE NOTED.
6. MOUNT DIFFUSER HORIZONTALLY. BLADE DEFLECTION ANGLE TO BE SET AT 45 ° FROM HORIZONTAL UNLESS OTHERWISE NOTED.



This item has been electronically signed and sealed by Michael Grose using a Digital Signature and date. Printed copies of this document are not considered signed and sealed and the signature must be verified on any electronic copies.

ISSUANCE

11/07/2025 PERMIT CYCLE 1

PROJECT NUMBER

7850130

PROJECT MANAGER

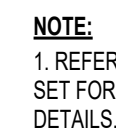
A. HENGSTEBECK

PROFESSIONAL

M. GROSE

DRAWN BY

J. MARTINEZ



- 3.2. TRADE CONTRACTOR RESPONSIBLE FOR NATURAL GAS PIPING INSTALLATION SHALL PROVIDE BMS GAS METER.



1. PROV



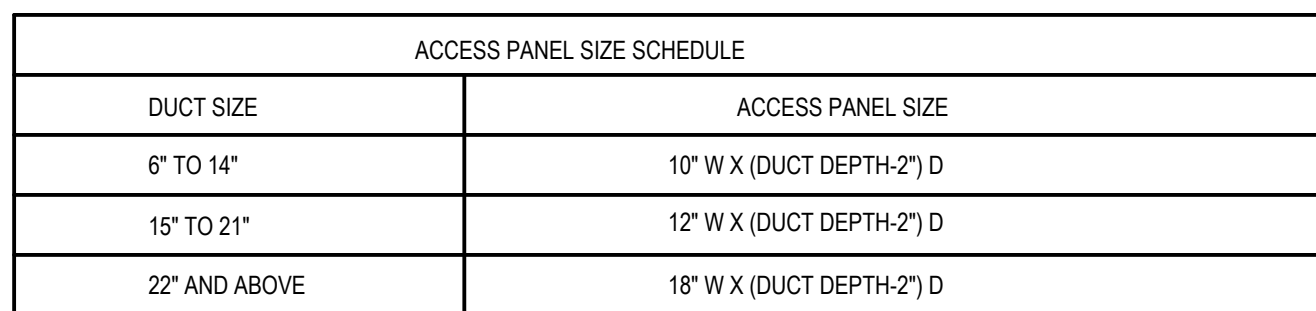
1. CONTRACTOR TO VERIFY THAT ENOUGH SPACE IS PROVIDED BETWEEN PIPE CLAMP AND PIPE.



A PIPE SUPPORT USED TO SUPPORT ROOF-MOUNTED GAS PIPES, ELECTRICAL CONDUIT, SOLAR PIPING AND OTHER MECHANICAL

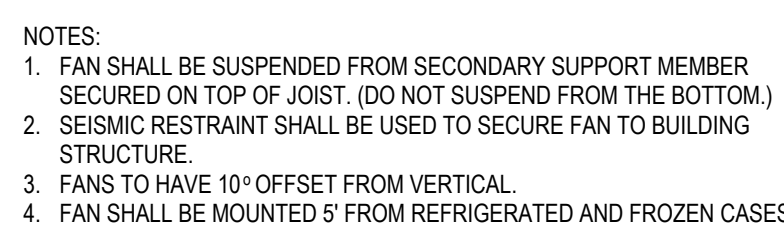
- 1-1/2" ID MAXIMUM
- PIPE CLEARANCE

- 22 PER CASE, 44 LBS. PER CASE

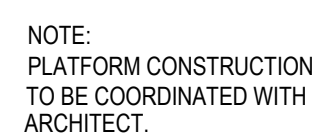


- NOTES:
1. PROVIDE FOUR CAM LATCHES ON PANELS LARGER THAN 18"x18" SIZE. REQUIRED AT ALL SMOKE DETECTORS, FIRE DAMPERS, SMOKE DAMPERS, COMBINATION FIRE/SMOKE DAMPERS, CONTROL DAMPERS AND HUMIDIFIERS ETC.
 2. ACCESS DOOR FOR FIRE DAMPERS, AND FIRE SMOKE DAMPERS SHALL BE LABELED.
 3. COORDINATE CEILING ACCESS PANEL REQUIREMENTS WITH ARCHITECT.

D6 DUCT ACCESS DOOR DETAIL
NOT TO SCALE



E6 **DESTRATIFICATION FAN DETAIL**
NOT TO SCALE



**ROOF MOUNTED CONDENSING UNIT
DETAIL**



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COMcheck Software Version COMcheckWeb
Mechanical Compliance Certificate

Project Information

Energy Code: 2015 IECC
Project Title: Lidl Norcross GA
Location: Atlanta, Georgia
Climate Type: 3a
Project Type: New Construction

Construction Site: 4990 JIMMY CARTER BLVD.
NORCROSS, Georgia 30093
Owner/Agent: Lidl US Operations LLC
3500 S. Clark St.
Arlington, Virginia 22202
Designer/Contractor: Progressive Architecture
Engineering & Planning I, LLC
1801 N. Graham Street
Charlotte, North Carolina 28206
704-731-8080

Additional Efficiency Package(s)

Credits: 1.0 Required 1.0 Proposed
Reduced Lighting Power, 1.0 credit

Mechanical Systems List

QuantitySystem Type & Description

- RTU-1 (Single Zone):
Heating: 1 each - Central Furnace, Gas, Capacity = 480 kBtu/h
Proposed Efficiency = 81.00% Et, Required Efficiency: 80.00 % Et
Cooling: 1 each - Single Package DX Unit, Capacity = 243 kBtu/h, Air-Cooled Condenser, Air Economizer
Proposed Efficiency = 12.00 EER, Required Efficiency = 9.80 EER
Proposed Part Load Efficiency = 15.40 IEER, Required Part Load Efficiency = 11.40 IEER
- RTU-2 (Single Zone):
Heating: 1 each - Central Furnace, Gas, Capacity = 480 kBtu/h
Proposed Efficiency = 81.00% Et, Required Efficiency: 80.00 % Et
Cooling: 1 each - Single Package DX Unit, Capacity = 221 kBtu/h, Air-Cooled Condenser, Air Economizer
Proposed Efficiency = 12.00 EER, Required Efficiency = 10.80 EER
Proposed Part Load Efficiency = 16.00 IEER, Required Part Load Efficiency = 12.20 IEER
- RTU-3 (Single Zone):
Heating: 1 each - Central Furnace, Gas, Capacity = 150 kBtu/h
Proposed Efficiency = 81.00% Et, Required Efficiency: 80.00 % Et or 78% AFUE
Cooling: 1 each - Single Package DX Unit, Capacity = 75 kBtu/h, Air-Cooled Condenser, Air Economizer
Proposed Efficiency = 12.20 EER, Required Efficiency = 11.00 EER
Proposed Part Load Efficiency = 17.30 IEER, Required Part Load Efficiency = 12.60 IEER
- RTU-4 (Single Zone):
Heating: 1 each - Central Furnace, Gas, Capacity = 180 kBtu/h
Proposed Efficiency = 81.00% Et, Required Efficiency: 80.00 % Et or 78% AFUE
Cooling: 1 each - Single Package DX Unit, Capacity = 126 kBtu/h, Air-Cooled Condenser, Air Economizer
Proposed Efficiency = 12.10 EER, Required Efficiency = 11.00 EER
Proposed Part Load Efficiency = 16.10 IEER, Required Part Load Efficiency = 12.60 IEER
- AC/CU-1 (Single Zone):
Split System Heat Pump
Heating Mode: Capacity = 24 kBtu/h,
Proposed Efficiency = 10.50 HSPF, Required Efficiency = 8.20 HSPF
Cooling Mode: Capacity = 16 kBtu/h,
Proposed Efficiency = 24.00 SEER, Required Efficiency = 14.00 SEER
Proposed Part Load Efficiency = 0.00 , Required Part Load Efficiency = 0.00
- EUH-1 (Unknown w/ PerimeterSystem):
Heating: 1 each - Unit Heater, Electric, Capacity = 27 kBtu/h
No minimum efficiency requirement applies

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Section # & Req.ID	Footing / Foundation Inspection	Complies?	Comments/Assumptions
C402.2.6 [FO12] ³	Radiant heating systems panels insulated to >=R-3.5 on face opposite space being heated.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	See the Envelope Assemblies table for values.
C403.2.4.5, C403.2.4.6 [FO9] ³	Snow/ice melting system sensors for future connection to controls. Freeze protection systems have automatic controls installed.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Exception: Requirement does not apply.

Additional Comments/Assumptions:

1 High Impact (Tier 1) 2 Medium Impact (Tier 2) 3 Low Impact (Tier 3)

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QuantitySystem Type & Description

- Water Heater:
Electric Storage Water Heater, Capacity: 50 gallons w/ Circulation Pump
No minimum efficiency requirement applies

Mechanical Compliance Statement

Compliance Statement: The proposed mechanical design represented in this document is consistent with the building plans, specifications, and other calculations submitted with this permit application. The proposed mechanical systems have been designed to meet the 2015 IECC requirements in COMcheck Version COMcheckWeb and to comply with any applicable mandatory requirements listed in the Inspection Checklist.

Joaquin Martinez Sabag Date: 2025.12.15 09:39:03-05'00'
Name - Title Signature Date 12/15/2025

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Section # & Req.ID	Plumbing Rough-In Inspection	Complies?	Comments/Assumptions
C404.5, C404.5.1, C404.5.2 [PL6] ³	Heated water supply piping conforms to pipe length and volume requirements. Refer to section details.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C404.6.1, C404.6.2 [PL3] ³	Automatic time switches installed to automatically switch off the recirculating hot-water system or heat trace.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C404.6.3 [PL7] ³	Pumps that circulate water between a heater and storage tank have controls that limit operation from startup to <= 5 minutes after end of heating cycle.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Exception: Requirement does not apply.
C404.7 [PL8] ³	Water distribution system that pumps water from a heated-water supply pipe back to the heated-water source through a cold-water supply pipe is a demand recirculation water system. Pumps within this system have controls that start the pump upon receiving a signal from the action of a user of a fixture or appliance and limits the temperature of the water entering the cold-water piping to 104°F.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.

Additional Comments/Assumptions:

1 High Impact (Tier 1) 2 Medium Impact (Tier 2) 3 Low Impact (Tier 3)

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COMcheck Software Version COMcheckWeb
Inspection Checklist

Energy Code: 2015 IECC

Requirements: 60.0% were addressed directly in the COMcheck software

Text in the "Comments/Assumptions" column is provided by the user in the COMcheck Requirements screen. For each requirement, the user certifies that a code requirement will be met and how that is documented, or that an exception is being claimed. Where compliance is itemized in a separate table, a reference to that table is provided.

Section # & Req.ID	Plan Review	Complies?	Comments/Assumptions
C103.2 [PR2] ¹	Plans, specifications, and/or calculations provide all information with which compliance can be determined for the mechanical systems and equipment and document where exceptions to the standard are claimed. Load calculations per acceptable engineering standards and handbooks.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C103.2 [PR3] ¹	Plans, specifications, and/or calculations provide all information with which compliance can be determined for the service water heating systems and equipment and document where exceptions to the standard are claimed. Hot water system sized per manufacturer's sizing guide.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C406 [PR9] ¹	Plans, specifications, and/or calculations provide all information with which compliance can be determined for the additional energy efficiency package options.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	

Additional Comments/Assumptions:

1 High Impact (Tier 1) 2 Medium Impact (Tier 2) 3 Low Impact (Tier 3)

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Section # & Req.ID	Mechanical Rough-In Inspection	Complies?	Comments/Assumptions
C402.2.6 [ME41] ³	Thermally ineffective panel surfaces of sensible heating panels have insulation >= R-3.5.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C403.2.12.1 [ME65] ³	HVAC fan systems at design conditions do not exceed allowable fan system motor nameplate hp or fan system bhp.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met. See the Mechanical Systems list for values.
C403.2.12.3 [ME117] ²	Fans have efficiency grade (FEG) >= 67. The total efficiency of the fan at the design point of operation <= 15% of maximum total efficiency of the fan.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Exception: Fans integral to equipment listed under Section C403.2.3.
C403.2.13 [ME71] ³	Unenclosed spaces that are heated use only radiant heat.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Exception: Requirement does not apply.
C403.2.4.7 [ME113] ²	Fault detection and diagnostics installed with air-cooled unitary DX units having economizers.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C403.2.6.1 [ME59] ¹	Demand control ventilation provided for spaces >500 ft2 and >25 people/1000 ft2 occupant density and served by systems with air side economizer, auto modulating outside air damper control, or design airflow >3,000 cfm.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C403.2.6.2 [ME115] ³	Enclosed parking garage ventilation has automatic contaminant detection and capacity to stage or modulate fans to 50% or less of design capacity.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Exception: Requirement does not apply.
C403.2.7 [ME57] ¹	Exhaust air energy recovery on systems meeting Table C403.2.7(1) and C403.2.7(2).	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Exception: Where the largest exhaust source is less than 75% of the design outdoor airflow.
C403.2.8 [ME116] ³	Kitchen exhaust systems comply with replacement air and conditioned supply air limitations, and satisfy hood rating requirements and maximum exhaust rate criteria.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C403.2.9 [ME60] ²	HVAC ducts and plenums insulated. Where ducts or plenums are installed in or under a slab, verification may need to occur during Foundation Inspection.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C403.2.9 [ME10] ²	Ducts and plenums sealed based on static pressure and location.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C403.2.9.1.3 [ME11] ³	Ductwork operating >3 in. water column requires air leakage testing.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Exception: Requirement does not apply.

1 High Impact (Tier 1) 2 Medium Impact (Tier 2) 3 Low Impact (Tier 3)

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This item has been electronically signed and sealed by Michael Grose using a Digital Signature and date. Printed copies of this document are not considered signed and sealed and the signature must be verified on any electronic copies.

ISSUANCE
1 11/07/2025 PERMIT CYCLE 1
1 12/18/2025 PERMIT CYCLE 02

PROJECT NUMBER
78650130
PROJECT MANAGER
A. HENGSTEBECK
PROFESSIONAL
M. GROSE
DRAWN BY
J. MARTINEZ
CHECKED BY
B. KOOMAN



This item has been electronically signed and sealed by Michel Grose using a Digital Signature and date. Printed copies of this document are not considered signed and sealed and the signature must be verified on any electronic copies.

ISSUANCE
1 11/07/2025 PERMIT CYCLE 1
1 12/18/2025 PERMIT CYCLE 02

PROJECT NUMBER
7855191
PROJECT MANAGER
A. HENGSTEBECK
PROFESSIONAL
M. GROSE
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MECHANICAL
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Section # & Req.ID	Mechanical Rough-In Inspection	Complies?	Comments/Assumptions
C403.3 [ME62] ¹	Air economizers provided where required, meet the requirements for design capacity, control signal, ventilation controls, high-limit shut-off, integrated economizer control, and provide a means to relieve excess outside air during operation.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C403.3.2 [ME16] ¹	Economizer operation will not increase heating energy use during normal operation.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C403.4.2.3.2.1 [ME121] ³	Closed-circuit cooling tower within heat pump loop have either automatic bypass valve or lower leakage positive closure dampers. Open-circuit tower within heat pump loop have automatic valve to bypass all heat pump water flow around the tower. Open- or closed-circuit cooling towers used in conjunction with a separate heat exchanger have heat loss by shutting down the circulation pump on the cooling tower loop.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Exception: Requirement does not apply.
C403.4.4.6 [ME110] ³	Multiple zone VAV systems with DDC of individual zone boxes have static pressure setpoint reset controls.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Exception: Requirement does not apply. <i>See the Mechanical Systems list for values.</i>
C408.2.2.1 [ME53] ³	Air outlets and zone terminal devices have means for air balancing.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C403.5, C403.5.1, C403.5.2 [ME123] ³	Refrigerated display cases, walk-in coolers or walk-in freezers served by remote compressors and remote condensers not located in a condensing unit, have fan-powered condensers that comply with Sections C403.5.1 and refrigeration compressor systems that comply with C403.5.2...	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.

Additional Comments/Assumptions:

1 High Impact (Tier 1) 2 Medium Impact (Tier 2) 3 Low Impact (Tier 3)

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Section # & Req.ID	Final Inspection	Complies?	Comments/Assumptions
C303.3, C408.2.5.3 [FI8] ³	Furnished O&M manuals for HVAC systems within 90 days of system acceptance.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C403.2.2 [FI27] ³	HVAC systems and equipment capacity does not exceed calculated loads.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C403.2.4.1 [FI47] ³	Heating and cooling to each zone is controlled by a thermostat control. Minimum one humidity control device per installed humidification/dehumidification system.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C403.2.4.1.1 [FI42] ³	Heat pump controls prevent supplemental electric resistance heat from coming on when not needed.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Exception: Requirement does not apply.
C403.2.4.1.2 [FI38] ³	Thermostatic controls have a 5 °F deadband.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C403.2.4.1.3 [FI20] ³	Temperature controls have setpoint overlap restrictions.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C403.2.4.2 [FI39] ³	Each zone equipped with setback controls using automatic time clock or programmable control system.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C403.2.4.2.1, C403.2.4.2.2 [FI40] ³	Automatic Controls: Setback to 55°F (heat) and 85°F (cool); 7-day clock, 2-hour occupant override, 10-hour backup	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C404.3 [FI11] ³	Heat traps installed on supply and discharge piping of non-circulating systems.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Exception: Requirement does not apply.
C404.4 [FI25] ²	All piping insulated in accordance with section details and Table C403.2.10.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C404.6.1 [FI12] ³	Controls are installed that limit the operation of a recirculation pump installed to maintain temperature of a storage tank. System return pipe is a dedicated return pipe or a cold water supply pipe.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C408.2.1 [FI28] ³	Commissioning plan developed by registered design professional or approved agency.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.

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Section # & Req.ID	Final Inspection	Complies?	Comments/Assumptions
C408.2.3.1 [FI31] ¹	HVAC equipment has been tested to ensure proper operation.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C408.2.3.2 [FI10] ¹	HVAC control systems have been tested to ensure proper operation, calibration and adjustment of controls.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C408.2.3.3 [FI32] ¹	Economizers have been tested to ensure proper operation.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C408.2.4 [FI29] ¹	Preliminary commissioning report completed and certified by registered design professional or approved agency.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C408.2.5.1 [FI7] ³	Furnished HVAC as-built drawings submitted within 90 days of system acceptance.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C408.2.5.3 [FI43] ¹	An air and/or hydronic system balancing report is provided for HVAC systems.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.
C408.2.5.4 [FI30] ¹	Final commissioning report due to building owner within 90 days of receipt of certificate of occupancy.	☐ Complies ☐ Does Not ☐ Not Observable ☐ Not Applicable	Requirement will be met.

Additional Comments/Assumptions:

1 High Impact (Tier 1) 2 Medium Impact (Tier 2) 3 Low Impact (Tier 3)

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